Map sketch

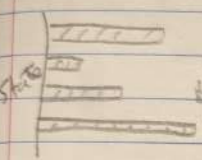
IDEAS

2 options

1. Choropleth area burned by state

2. Symbol Map fire locations, size = area.

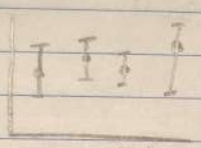
This is Aus



State

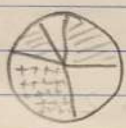
Avg burn time

Area burnt



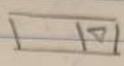
year / state

Ingnition vs extinguish time

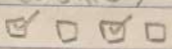


Ignition source per fire

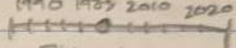
Filter



State selection or fire type (bushfire)



Filter by cause



1990 1985 2010 2020

Filter by year

Australian Bushfires - visual analysis

Harkirat Singh

Sheet 1

Planning

CATEGORISE

→ Geographic view

The map = are the fires getting worse? - Link to climate

→ Temporal view

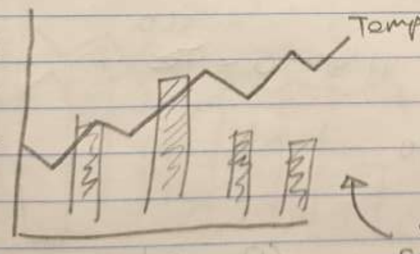
Trends over time - Are fired getting worse - Hotspots

→ Categorical view:

Breakdown - what states are most effective?

- what are the causes

Combine and Refine

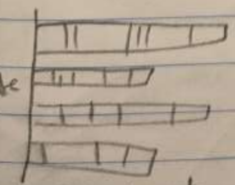


area burnt

Temp

years

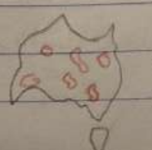
combine fire area and climate to show correlation



State

area burnt

Refined chart to have stacked charts to show yearly contribution for each state



Use and overlay actual fires.

Must have accurate.

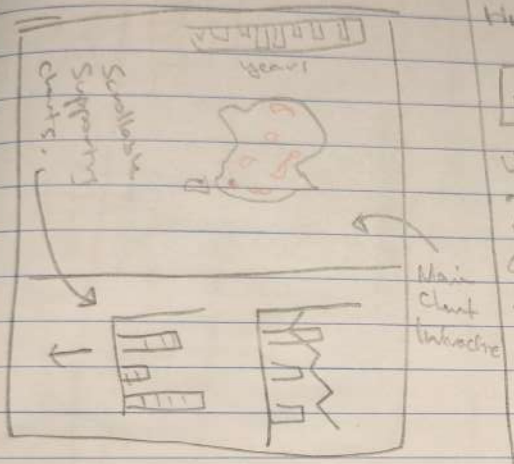
QUESTION

1. Does this vis tell a clear story

2. To answer
   - Is there a data driven link between hotter years to worse fires
   - What are the primary causes
   - User explore specific events.

LAYOUT

Aus bushfires



OPERATION

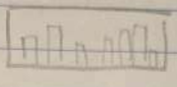
Australian Bushfires

Harkirat Singh

Sheet 2

Layout and Interaction

- Filter by time

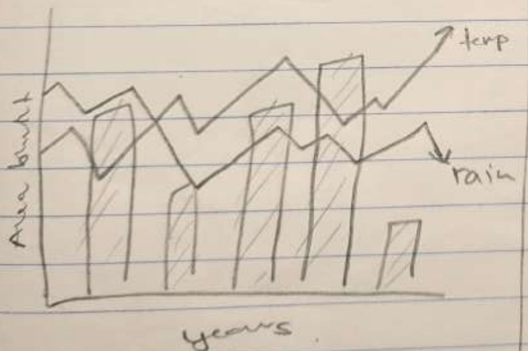


User brushes an area that is moveable.
Cross filters all charts

- Highlighting

Hovering over any fire will highlight it and show a tooltip with relevant specific info

Focus.



Primary focus is the correlation between climate drivers and fire extent. The map provides the "where" context. Could also look into "who" has been affected

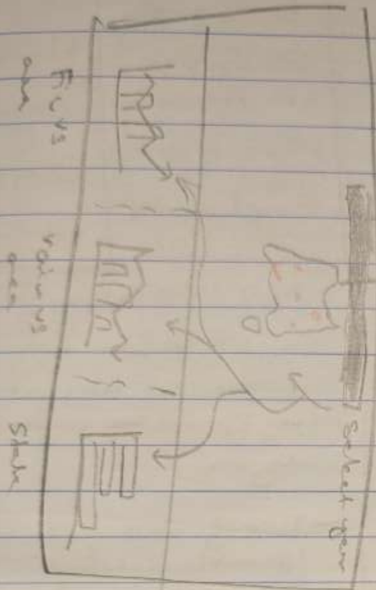
Discussion

- Need to decide between a single slider or brushable timeline for year selection

- How to show temp and rain without being cluttered
Possibly move to separate graphs?

- Have into normalisation. which will be more powerful to show relative area burnt

LAYOUT



OPERATIONS

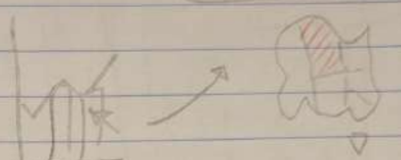
year slider to select a specific year.

Australian Bushfires

Harkirat Singh

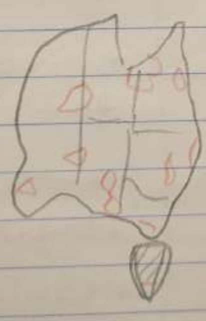
Sheet 3

Refined dashboard.



Revolve around main chart to show visual representation of the data on the map

Focus



→ Interactive map.
able to explore geographic patterns over the years.
"What": If a user sees a bad fire year on a chart then looks at the map. It will provide the complete scale of damage.
"Where" (map) to "Why" (climate) to "who?" (stats)

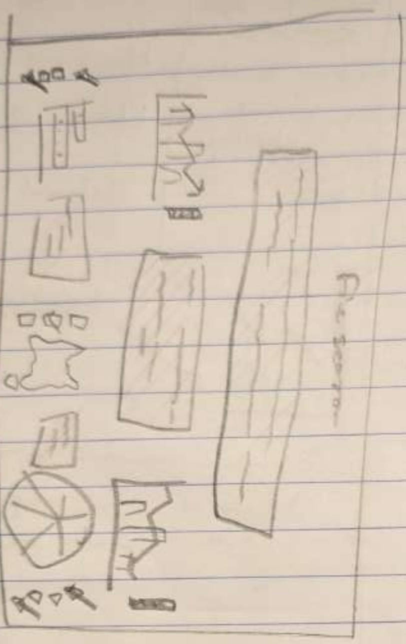
Discussion

→ Layout is strong because it creates a clear visual hierarchy. Map first details second.

→ The slider will only select ONE year. Is that too restrictive. will a brush be more suited to allow multiple year selection

→ Colour palette selection → red and grey/black themes.

LAYOUT



OPERATIONS

Each graph has its own set of controls when interacting is most likelihood

Australian Bushfire

Heatmap

Single

Stack 4

Alternate

- Map Interaction
  Simplified to only accommodate key events that would be impactful to analysis.

- Hovering still present to provide detailed specifics

Focus.

- Guided narrative

· Not exactly a story but more of an open ended tool. The focus is to read the analysis in a defined order

- More focus on presentation and the exploration

Discussion

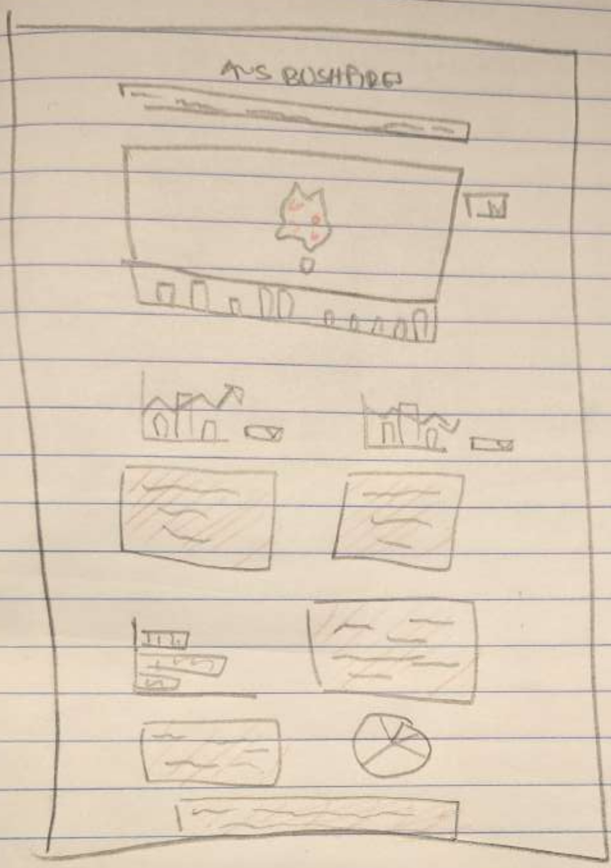
+ very strong narrative flow

+ more detailed explanation along each specific chart

+ Simpler interaction

— no cross filtering so users can't compare charts easily.

LAYOUT



Australia
Bushfires
Harkirat
Singh
Sheet 5
Final design

FINAL OPERATIONS

Primary interactions is the year brush slider.

Highlight specific years.

Comprehensive tooltips on all marks for specific quantitative data

Focus

→ Balanced, multipurpose perspective high level geographic view with deep multi-faceted analytical charts.

→ Key design is using local interactions to simplify filtering process and not always navigating back to the one universal filter

Discussion

→ Superior to the story telling perspective

→ Fragments data over a larger page

→ User driven exploration and side panels provide readability and space